

Appendix 4. Notes for Register R/W

- A secure bus master issues a secure access using an address marked as secure by IDAU/SAU or MSAU
- A secure bus master issues a non-secure access using an address marked as non-secure by IDAU/SAU or MSAU
- A non-secure bus master issues a non-secure access using an address marked as non-secure by IDAU/SAU or MSAU.

Table A4.1 Type of register notes (S-TYPE)

TYPE	UM description
S-TYPE-1	Only Secure access can write to this register. Read access is always allowed. Non-secure write access is ignored, but TrustZone access error is not generated.
S-TYPE-2	Read access is always allowed If the security attribution is configured as Secure: • Secure write access is allowed • Non-secure write access is ignored, but TrustZone access error is not generated.
	If the security attribution is configured as Non-secure: • Secure write access is ignored, but TrustZone access error is not generated • Non-secure access is allowed.
S-TYPE-3	If the security attribution is configured as Secure: • Secure access is allowed • Non-secure write access is ignored and Non-secure read access is read as 0, TrustZone access error is generated.
	If the security attribution is configured as Non-secure: • Secure write access is ignored and Secure read access is read as 0, TrustZone access error is generated • Non-secure access is allowed.
S-TYPE-4	If the security attribution is configured as Secure: • Secure access is allowed • Non-secure write access is ignored and Non-secure read access is read as 0, but TrustZone access error is not generated.
	If the security attribution is configured as Non-secure: • Secure write access is ignored and Secure read access is read as 0, but TrustZone access error is not generated • Non-secure access is allowed.
S-TYPE-5	Access is always allowed.
S-TYPE-6	Secure access is allowed. Non-secure write access is ignored, and Non-secure read access is read as 0, TrustZone access error is generated.
S-TYPE-7	Secure write access is ignored, and Secure read access is read as 0, TrustZone access error is generated. Non-secure access is allowed.

Note: A non-secure bus master does NOT issue any access using an address marked as secure by IDAU/SAU or MSAU.

Table A4.2 Type of register notes (P-TYPE)

TYPE	UM description
P-TYPE-1	Privileged write access is allowed. Read access is always allowed. Unprivileged write access is ignored, but TrustZone access error is not generated.
P-TYPE-2	Privileged access is allowed. Unprivileged write access is ignored, and Unprivileged read access is read as 0, TrustZone access error is generated.
P-TYPE-3	If the privilege attribution is configured as Privileged: • Privileged access is allowed • Unprivileged write access is ignored and Unprivileged read access is read as 0, TrustZone access error is generated.
	If the privilege attribution is configured as Unprivilege: • Privileged access and Unprivileged access are allowed.
P-TYPE-4	If the privilege attribution is configured as Privileged: • Privileged access is allowed • Unprivileged write access is ignored and Unprivileged read access is read as 0, TrustZone access error is not generated.
	If the privilege attribution is configured as Unprivilege: • Privileged access and Unprivileged access are allowed.
P-TYPE-5	Access is always allowed.